Indirect Auto Lending Needs a Smarter Allocation Strategy

Prescriptive Analytics Final Project | Shelia Bond | April 2026

The Decision

The VP of Consumer Lending must decide how to allocate $25M quarterly indirect auto budget across 10 active dealerships — maximizing net yield while staying within approved risk limits and loan growth targets.

The Problem

Without a systematic approach, allocation defaults to habit or relationship — not performance. Dealers vary widely in yield potential and risk profile, making the choice of who gets how much consequential.

Why It Matters

Poor allocation reduces yield and may breach the portfolio risk threshold — directly impacting the credit union's financial performance.

The Model Maximizes Yield Within Four Hard Constraints

Component	Detail
Decision Variable	Dollar amount allocated to each of 10 dealers per quarter (continuous)
Objective	Maximize total net yield across the full portfolio
CONSTRAINTS	
Budget Floor	Total allocation must be at least $18M per quarter (loan growth target)
Budget Ceiling	Total allocation must not exceed $25M per quarter
Portfolio Risk Limit	Weighted average portfolio risk score must stay below 2.8
Dealer Concentration Cap	No single dealer may receive more than $3M per quarter
KEY TRADEOFF	
Yield vs. Risk	Higher yield comes with higher risk — maximizing return pushes the portfolio toward riskier dealers

Model type: Mixed-Integer Linear Program (MILP) | Solver: PuLP/CBC

9 of 10 Dealers Receive Funding — BMW Excluded for Lowest Yield

$25M

Capital Deployed

9 of 10

Dealers Funded

⚠ BMW Dealer Not Allocated

Lowest yield (4.08%) with moderate risk (2.25) — no competitive advantage over funded dealers

Dealer	Yield Rate	Risk Score	Optimal Allocation ($M)
Mercedes Dealer	7.88%	3.40	$3.00
Nissan Dealer	7.33%	2.96	$3.00
Ford Dealer	6.10%	3.23	$3.00
Honda Dealer	5.73%	2.70	$3.00
Acura Dealer	5.22%	1.62	$3.00
Kia Dealer	5.16%	2.92	$3.00
Toyota Dealer	4.85%	2.70	$3.00
Dodge Dealer	4.73%	1.81	$1.00
Mazda Dealer	4.73%	1.81	$3.00
BMW Dealer	4.08%	2.25	—
Portfolio Total	**5.83%**	**2.63**	**$25.00**

Risk Limit and Dealer Caps Drive Outcomes — Growth Target Does Not

Parameter	Scenario Tested	Net Yield Impact	Verdict
Growth Floor (min $18M)	±20% change	< 5% swing	**Robust ✓**
Portfolio Risk Limit (2.8)	±20% change	Significant shift	**Fragile ⚠**
Dealer Cap ($3M max)	±20% change	Significant shift	**Fragile ⚠**

What This Means: Growth Floor

The VP can adjust the minimum volume target by ±20% without meaningfully changing which dealers are funded or what the portfolio yields. This constraint is not the binding factor.

What This Means: Risk Limit & Dealer Cap

These are the two levers that actually control outcomes. Tightening the risk limit forces out higher-yield dealers; reducing the dealer cap forces reallocation. Both deserve scrutiny before acting.

Time Is Not a Factor Now — But Would Be Under These Conditions

The current model is a static, single-period decision. Time would become highly relevant if any of the following apply:

1

Changing Dealer Performance

Yield rates or risk scores may shift due to market cycles or dealer-specific changes — requiring a dynamic model to re-optimize allocations over time.

2

Budget Fluctuations

If the quarterly budget varies or unallocated funds roll over, a time dimension allows the model to plan across periods and smooth allocations.

3

Seasonality

If demand and risk follow seasonal patterns, the model could shift allocations toward peak demand periods or away from higher-risk windows.

Validate Nonlinear Effects Before Full Deployment

Where the Model Has Limits

Yield is not linear
Pushing volume to a dealer may exhaust high-quality borrowers, reducing marginal yield.

Risk is not fixed
Forcing volume growth may cause a dealer to loosen underwriting, raising actual risk scores.

What We Should Do

Recommendation
Do not fully deploy the model's output yet. Prioritize validation first.

Key Caveat
The recommendation is fragile to risk limit and dealer cap assumptions — both are actively managed parameters.

First Step
Run a pilot with a subset of dealers to measure real yield/risk response to volume changes.

Bottom line: The model's logic is sound. The inputs need real-world validation before the VP acts on the allocation.